Renal Tumors & Urinary Bladder Pathology

Pathology GUE Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty. Attempt every question before checking the answer key on the last page.

4 Easy **10 Medium** **6 Hard**

1. Irrespective of size, which renal tumor should always be regarded as potentially malignant? EASY

A. Angiomyolipoma
B. Oncocytoma
C. Renal papillary adenoma
D. Wilms tumor
E. Urothelial papilloma

2. Which benign renal tumor is strongly associated with tuberous sclerosis and carries a risk of spontaneous hemorrhage? EASY

A. Oncocytoma
B. Angiomyolipoma
C. Papillary adenoma
D. Clear cell RCC
E. Collecting duct RCC

3. What is the most common subtype of renal cell carcinoma? EASY

A. Papillary RCC
B. Chromophobe RCC
C. Clear cell RCC
D. Collecting duct RCC
E. Medullary RCC

4. Bladder diverticula are most commonly caused by which acquired condition? EASY

A. Urinary schistosomiasis
B. Prostatic enlargement
C. Tuberculosis
D. Chemotherapy exposure
E. Urachal remnant patency

5. A 55-year-old presents with secondary polycythemia and a renal mass. Which renal tumor is classically associated with this paraneoplastic finding via ectopic erythropoietin production? MEDIUM CASE

A. Angiomyolipoma
B. Oncocytoma
C. Renal cell carcinoma
D. Wilms tumor
E. Urothelial carcinoma

6. Which renal tumor histologically closely mimics oncocytoma, sharing the same cell of origin (intercalated cells of the collecting duct) but carrying malignant potential, albeit with excellent prognosis? MEDIUM

A. Clear cell RCC
B. Papillary RCC
C. Chromophobe RCC
D. Collecting duct RCC
E. Angiomyolipoma

7. Von Hippel-Lindau disease is associated with all of the following EXCEPT: MEDIUM

A. Hemangioblastoma
B. Pheochromocytoma
C. Renal cell carcinoma
D. Embryonal rhabdomyosarcoma
E. Papillary cystadenoma of epididymis

8. A 3-year-old presents with an abdominal mass. Biopsy shows blastemal, epithelial, and stromal components with abortive glomeruli and tubules. Which gene on chromosome 11p13 is implicated? MEDIUM CASE

A. VHL
B. WT1
C. TSC1
D. TFE3
E. PKD1

9. Clear cell renal cell carcinoma is characterized by which chromosomal abnormality? MEDIUM

A. Trisomy 7 and 17
B. Loss of sequences on chromosome 3p
C. Xp11.2 translocation
D. Monosomy of multiple chromosomes
E. 11p13 deletion

10. Which statement about bladder diverticula is correct? MEDIUM

A. Congenital diverticula are more common than acquired
B. Acquired diverticula are mainly due to prostatic hyperplasia or neoplasia
C. Diverticula never predispose to carcinoma
D. Diverticula are exclusively located in the bladder dome
E. Diverticula always require urgent surgery

11. A renal transplant recipient develops soft, yellow bladder mucosal plaques. Biopsy shows macrophages with abundant granular cytoplasm and laminated, calcified concretions both within and between the cells. What is the diagnosis? MEDIUM CASE

A. Interstitial cystitis
B. Polypoid cystitis
C. Malakoplakia
D. Carcinoma in situ
E. Papillary urothelial carcinoma

12. Which histologic feature distinguishes low-grade from high-grade papillary urothelial carcinoma? MEDIUM

A. Low-grade cells are discohesive with loss of polarity; high-grade cells remain cohesive
B. Low-grade cells maintain polarity and show mild-moderate atypia; high-grade cells are discohesive with severe atypia and necrosis
C. Low-grade tumors always invade muscularis propria; high-grade tumors never invade
D. Low-grade lesions lack an umbrella cell layer; high-grade lesions retain it
E. There is no histologic distinction, only clinical behavior differs

13. A woman with chronic pelvic pain undergoes cystoscopy showing fissures and punctate hemorrhages (glomerulations) in the bladder mucosa, with inflammation and fibrosis of all bladder wall layers. What is the main purpose of biopsy in this condition? MEDIUM CASE

A. To confirm malakoplakia
B. To rule out carcinoma in situ
C. To diagnose schistosomiasis
D. To grade a papillary tumor
E. To assess for Wilms tumor

14. What is the key histologic difference between urothelial papilloma and PUNLMP? MEDIUM

A. Papilloma shows severe atypia; PUNLMP does not
B. PUNLMP has thicker urothelium (>8 cell layers) than papilloma (7-8 layers)
C. Papilloma is always invasive; PUNLMP is not
D. PUNLMP lacks a fibrovascular core
E. Papilloma occurs only in elderly patients

15. A 60-year-old smoker presents with painless hematuria. Cystoscopy and biopsy show malignant urothelial cells confined to flat epithelium with no basement membrane invasion, ranging from full-thickness involvement to scattered cells (pagetoid spread). What is the correct AJCC/UICC stage? HARD CASE

A. Ta
B. Tis
C. T1
D. T2
E. T3a

16. Regarding renal cell carcinoma subtypes, which pairing of subtype and distinguishing feature is INCORRECT? HARD

A. Clear cell RCC — loss of chromosome 3p sequences
B. Papillary RCC — can be multifocal and bilateral
C. Chromophobe RCC — hobnail pattern epithelium in fibrotic stroma
D. Collecting duct RCC — accounts for less than 1% of RCC
E. Clear cell RCC — frequently invades renal vein or IVC

17. A bladder biopsy shows tumor cells extending microscopically beyond the bladder wall into perivesical fat, without gross extravesical involvement. What AJCC/UICC T-stage is assigned? HARD CASE

A. T1
B. T2
C. T3a
D. T3b
E. T4

18. Which of the following correctly matches a bladder sarcoma with its typical age group? HARD

A. Leiomyosarcoma — most common bladder sarcoma in infancy/childhood
B. Embryonal rhabdomyosarcoma — most common bladder sarcoma in infancy/childhood
C. Embryonal rhabdomyosarcoma — most common bladder sarcoma in adults
D. Leiomyoma — most common malignant bladder sarcoma overall
E. Leiomyosarcoma — exclusively seen in pediatric patients

19. A patient from a schistosomiasis-endemic region develops a bladder malignancy with squamous differentiation. A separate patient has a tumor arising from a patent urachal remnant. Which histologic subtypes are respectively most likely in these two patients? HARD CASE

A. Urothelial carcinoma; squamous cell carcinoma
B. Squamous cell carcinoma; adenocarcinoma
C. Adenocarcinoma; squamous cell carcinoma
D. Leiomyosarcoma; urothelial carcinoma
E. Squamous cell carcinoma; leiomyosarcoma

20. Which statement correctly differentiates the genetics of Wilms tumor from the genetics of clear cell RCC? HARD

A. Wilms tumor involves WT1 at 11p13; clear cell RCC involves loss of chromosome 3p sequences
B. Wilms tumor involves TSC1/TSC2; clear cell RCC involves WT1
C. Both involve the same VHL gene mutation
D. Wilms tumor involves 3p deletion; clear cell RCC involves 11p13 deletion
E. Neither tumor has an identified genetic association

A Answer key & rationale

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**C**	All renal papillary adenomas are regarded as potentially malignant irrespective of size (>1 cm is indicative of malignancy).
2	**B**	Angiomyolipoma occurs in 25-50% of tuberous sclerosis patients and risks spontaneous hemorrhage due to large size.
3	**C**	Clear cell RCC is the most common RCC subtype.
4	**B**	Acquired bladder diverticula are mainly due to prostatic enlargement (hyperplasia or neoplasia).
5	**C**	Renal cell carcinoma causes paraneoplastic secondary polycythemia via ectopic erythropoietin production.
6	**C**	Chromophobe RCC closely mimics oncocytoma, arising from the same intercalated collecting-duct cells, but has excellent prognosis despite malignancy.
7	**D**	VHL disease is associated with hemangioblastoma, pheochromocytoma, RCC, and papillary cystadenoma of epididymis/broad ligament — not rhabdomyosarcoma.
8	**B**	WT1 at chromosome 11p13 is the gene implicated in Wilms tumor.
9	**B**	Clear cell RCC shows loss of sequences on the short arm of chromosome 3 (3p).
10	**B**	Acquired bladder diverticula are mainly due to prostatic enlargement (hyperplasia or neoplasia); congenital ones are less common.
11	**C**	Macrophages with granular cytoplasm and laminated calcified Michaelis-Gutmann bodies define malakoplakia.
12	**B**	Low-grade papillary UC cells maintain polarity with mild-moderate atypia; high-grade cells are discohesive with severe atypia, pleomorphism, and necrosis.
13	**B**	The main role of biopsy in interstitial cystitis (Hunner ulcer) is to rule out carcinoma in situ.
14	**B**	PUNLMP is distinguished from papilloma mainly by thicker urothelium (>8 cell layers vs. 7-8 in papilloma).
15	**B**	Malignant cells confined to flat urothelium without basement membrane invasion defines carcinoma in situ, staged as Tis.
16	**C**	A hobnail pattern within fibrotic stroma describes collecting duct (Bellini duct) RCC, not chromophobe RCC (which shows eosinophilic granular cells with perinuclear halo).
17	**C**	Microscopic extravesical invasion (beyond the bladder wall into perivesical fat without gross involvement) is staged as T3a.
18	**B**	Embryonal rhabdomyosarcoma is the most common sarcoma of the bladder in infancy/childhood; leiomyosarcoma is most common in adults.
19	**B**	Schistosomiasis-associated bladder cancer is typically squamous cell carcinoma; tumors arising from urachal remnants are typically adenocarcinoma.
20	**A**	Wilms tumor is driven by WT1 (11p13) [and WT2 at 11p15]; clear cell RCC is characterized by loss of chromosome 3p sequences.